

Intrinsic Value of Void Grubs:

Free Scrap Pricing, Opportunity Gold, and Contest Rationality

Abstract

We separate two quantities often sold as one: the *free scrap* from taking all three Voidgrubs with the fight held fixed (local gold, river XP, Touch of the Void / Hunger mite) versus the *take-regime association* of ending 3-0 after early-state controls. Map-WR effects are reported in percentage points (pp) rounded to two decimals. Sample: $n = 11,168$ professional maps (patch ≥ 15.09 or year 2026; grub_sum = 3; 2025-05-02 to 2026-07-18). We price the scrap through gold@10 / xp@10 logits, then embed leave-window *farm gold* (waves and jungle) and *teamfight kill-gold nets* in the contest rule.

Results. At even gold@10, +90g maps to +1.75 pp ($\approx +1.95$ pp per 100g). Preferred scrap (90g local + 8s melee Touch $\rightarrow$ first outer plate) is +1.92 pp; joint gold+XP +3.37 pp; upper (joint + 20s burn) +3.79 pp. Preferred leave-window farm is 340g $\rightarrow$ +6.56 pp. Median TF net +2 / -2 kills (300g bounty) map to +11.40 pp / -11.60 pp. Under the preferred contest model, at $p = 0.25$ the edge versus leave is -11.45 pp ($p^* \approx 0.68$). **LEAVE / do nothing.**

1 Introduction

Void grubs spawn near minute eight. Discourse treats them as scrap not worth a bad fight, or as a must-contest tempo objective. Large-sample take-regime win rates and practice-tool mechanical breakdowns both look empirical; they measure different quantities.

Two July 2026 debates fix the estimand. Against a Twitter analyst, LS starts from 30g and 65 XP per grub and sandbox isolation of Touch of the Void. Against a tier-2 coach (Furia vs G2, ~7:30), the claim is probability-bar arithmetic: if free grubs move win probability only slightly, a 25/75 river fight to deny that shift is irrational; farming waves and camps (“doing nothing”) is an active macro.

2 Estimands

Let $T = \{90\text{g local, } 195 \text{ XP in } 2000\text{-unit range, Touch at 3 stacks}\}$ and $Y \in \{0, 1\}$ map win. The scrap value (what the camp buys if nobody has to die for it) is

$$\Delta_{\text{obj}} \equiv \mathbb{E}[Y \mid \text{state}, T] - \mathbb{E}[Y \mid \text{state}, \neg T] \quad (\text{combat lottery held fixed}). \quad (1)$$

We approximate (1) by mapping components through gold@10 / xp@10 logits fit on the era sample.

Take-regime association (non-target) is the unique effect of ending blue 3-0 after gold/kills/xp@10 controls. It encodes selection and conversion, not the tooltip of T .

2.1 Contest decision with farm and kill gold

During a ~60-90s window, leave captures farm gold F (waves + jungle) and gifts T ; contest forgoes F and enters a fight with win probability p . Let K_{win} (resp. K_{lose}) be map-WR points from net kill gold on a won (lost) fight, and

let $L = K_{\text{lose}} + \Delta_{\text{obj}}$ be full loss including the gift. Then

$$\text{EV}_{\text{leave}} = -\Delta_{\text{obj}} + F, \quad (2)$$

$$\text{EV}_{\text{contest}}(p) = p(\Delta_{\text{obj}} + K_{\text{win}}) + (1 - p)(-L), \quad (3)$$

$$\text{Edge}(p) = \text{EV}_{\text{contest}}(p) - \text{EV}_{\text{leave}}, \quad (4)$$

$$p^* = \frac{L - \Delta_{\text{obj}} + F}{L + \Delta_{\text{obj}} + K_{\text{win}}}. \quad (5)$$

Headline priors: F = two laner waves + one jungle camp; median TF net +2 kills on win and -2 on loss at 300g early bounty.

3 Data and methods

Oracle’s Elixir professional maps; era filter patch ≥ 15.09 or year 2026; grub_sum = 3; $n = 11,168$. OE has no fight logs; farm and kill packages are structural sensitivity, mapped through the same gold@10 logit

$$\text{logit } P(Y = 1) = \beta_0 + \beta_1 \cdot \text{golddiff}_{10}, \quad (6)$$

with $\beta_0 = 0.0745$, $\beta_1 = 0.000782$, $n_{\text{fit}} = 9,786$.

Touch of the Void at three stacks (melee, pre-26.11): 12 true / 0.5s for 4s per AA refresh (96 true / cycle). At three stacks, Hunger of the Void also summons one Voidmite on structure combat (15s CD) that applies/refreshes Touch. Burn $\rightarrow$ plate gold uses current wiki Turret Plating: 120g per plate; outer turret 9000 HP; first plate band 100% $\rightarrow$ 70% (2700 HP) — the relevant band for an early grub $\rightarrow$ push. XP is 65 per grub in a 2000-unit radius, shared equally among alive allies in range; team xpdiff@10 still rises by the full 195 sum (solo jungler soak vs duo/trio changes who spikes a level). Patch 26.11 increases Touch tick damage; the post-buff 20s siege bound (and Hunger-mite refresh) enters the upper sensitivity.

4 Results

4.1 Intrinsic package bounds

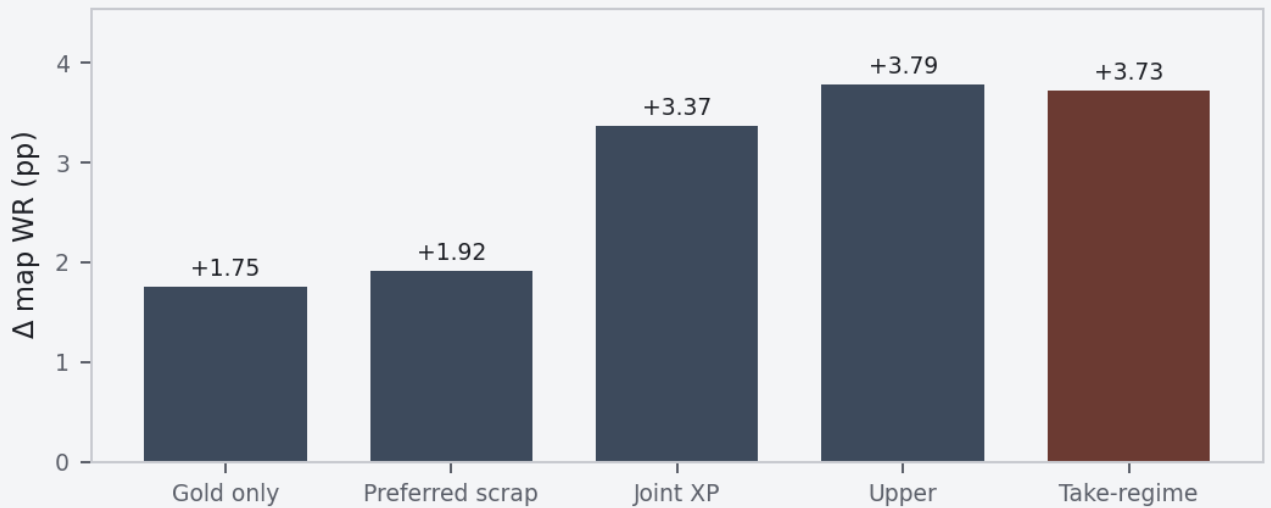


Figure 1. Intrinsic $\Delta \text{ map WR (pp)}$ at even versus contaminated take-regime association (last bar).

Bound	pp	Contents
Lower	+1.75	90g local only
Preferred scrap	+1.92	90g + 8s Touch→1st plate
Joint gold+XP	+3.37	+195 team XP (river share)
Upper	+3.79	joint + 20s Touch
LS mid burn alone	+0.20	1.5 AA on full turret
Take-regime (not scrap)	+3.73	blue 3-0 controls@10

Table 1. Intrinsic bounds (display pp to two decimals).

Gold slope at even: +1.95 pp per 100g. Brief 8s Touch: +0.17 pp; 20s melee Touch: +0.42 pp.

4.2 Leave-window farm gold (waves and jungle)

Leave-window package	Gold	WR pp
1 laner clears one wave	120	+2.34
2 laners each clear one wave + jungler one camp	340	+6.56
3 waves + 2 camps	560	+10.67

Table 2. Structural farm packages during the contest window.

4.3 Teamfight kill-gold nets

Net kills (contester)	Gold	WR pp
-4	-1200	-22.20
-3	-900	-17.08
-2	-600	-11.60
-1	-300	-5.85
+0	0	+0.00
+1	300	+5.80
+2	600	+11.40
+3	900	+16.66
+4	1200	+21.49

Table 3. Net kill differential $\times 300g$ early bounty, mapped at even.

4.4 Contest EV

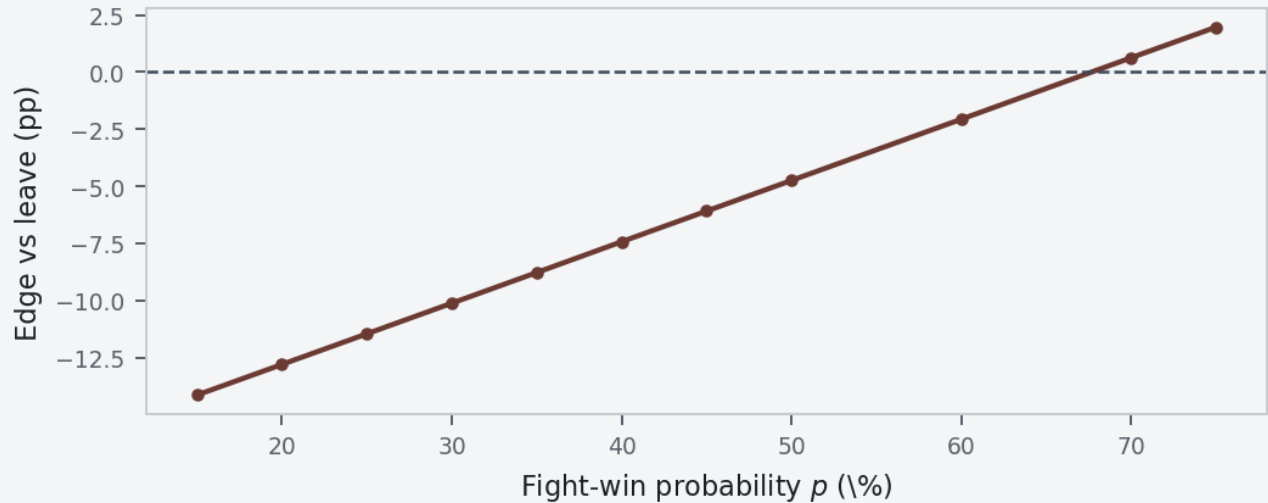


Figure 2. Edge of contesting minus leaving under preferred farm and median ± 2 kill nets. $\Delta_{\text{obj}} = +1.92$, $F = +6.56$, $K_{\text{win}} = +11.40$, $L = +13.52$, $p^* = 0.68$.

p	EV contest (pp)	Edge vs leave (pp)	Verdict
0.15	-9.49	-14.13	AVOID
0.20	-8.15	-12.79	AVOID
0.25	-6.81	-11.45	AVOID
0.30	-5.47	-10.11	AVOID
0.35	-4.13	-8.77	AVOID
0.40	-2.78	-7.42	AVOID
0.45	-1.44	-6.08	AVOID
0.50	-0.10	-4.74	AVOID
0.60	+2.58	-2.06	AVOID
0.70	+5.27	+0.63	CONTEST
0.75	+6.61	+1.97	CONTEST

Table 4. Preferred contest schedule (farm + median ± 2 kills).

Without farm or kill bonus (baseline), leave EV is only $-\Delta_{\text{obj}}$; at $p = 0.25$ edge = -7.74 pp, $p^* = 0.75$.

Scenario	Δ_{obj}	F	K_{win}	L	p^*	Edge _{0.25}
No farm, no kill bonus	+1.92	+0.00	+0.00	+13.52	0.75	-7.74
Preferred farm, no kill bonus	+1.92	+6.56	+0.00	+13.52	1.18	-14.30
Preferred farm + median ± 2 kills	+1.92	+6.56	+11.40	+13.52	0.68	-11.45
High farm + median ± 2 kills	+1.92	+10.67	+11.40	+13.52	0.83	-15.56
Preferred farm, -3 kill loss	+1.92	+6.56	+11.40	+19.00	0.73	-15.56

Table 5. Decision sensitivity (pp).

5 Discussion

Is LS right? On the Furia *decision* — do not take a 25/75 river fight to deny this camp — yes. On literal “0.10 pp”, no: that is rhetorical undersell. The scrap is a couple of percentage points of map WR (+1.92 pp preferred), not a tenth of one, and still tiny next to leave-window farm (+6.56 pp) or dying twice in river (-11.60 pp). Citing +3.73 pp take-regime association as “grub value” switches estimands: that number is about teams that already converted a take, not about 30g / 65 XP / Touch in isolation.

6 Limitations

- Resource $\rightarrow$ WR logits are associational, not experimental 90g grants.
- River XP is local (radius 2000, shared); OE team xpdiff still counts the full 195 sum — who gets the levels (solo jungler vs trio) is not modeled beyond that accounting.
- Burn $\rightarrow$ gold uses wiki first-plate economics (120g / 2700 HP); real sieges are path-dependent, and Hunger-mite timing varies.
- Farm packages are structural (window gold), not OE-timed pathing logs.
- Kill nets use a flat 300g early bounty; assist splits and shutdowns are ignored.
- No professional fight logs; median ± 2 kills is a stated prior, not a filmed distribution.
- Gold@10 is post-spawn.
- Patch 26.11 buffs Touch; post-buff 20s siege / Hunger-mite bounds are upper sensitivity only.
- One objective and one fight lottery; not a full pathing equilibrium.

7 Conclusion

What is the scrap? Everything you get from taking all three Voidgrubs without needing the fight to go your way: 90g local gold, 195 XP in river range, and Touch of the Void (plus Hunger mite at three stacks). Converted into map win rate, that scrap is about +1.92 pp — not “free winrate,” and not LS’s rhetorical 0.10 pp either.

Should you contest a bad river? If you leave, you gift that scrap but you keep farming waves/camps (+6.56 pp) and you avoid a teamfight where a median -2 kill swing is already -11.60 pp. Contesting at $p = 0.25$ is still -11.45 pp worse than leaving ($p^* \approx 0.68$). Default to leave when the river is clearly unfavored; walk up when you honestly win the fight often enough to clear that bar, or when the camp is free.